Phase Transitions of Thermoresponsive Polymers

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Introduction:

- Investigation of phase transition dynamics in thermoresponsive polymers using a non-isothermal model.
- Energy conservation in simulations accurately represents the transient behavior of polymer precipitation and associated energy changes during phase transitions.
- Detailed insights into the process of thermally induced phase transitions are obtained through simulations.
- Identification of each polymer's radius of gyration that controls the dynamics of phase transitions.
- Used three distinct concentrations within a hydrogel bead.

Why Thermoresponsive Polymers (TRPs):

- Allows drugs to be delivered in a controlled manner.
- 2 types:
  - Lower Critical Solution Temperature (LCST)
  - Upper Critical Solution Temperature (UCST)

poly(N-isopropylacrylamide) (PNIPAM) is one of many LCST TRPs.

- PNIPAM is used in medical applications due to its LCST around 32 °C which is between room temperature and body temperature.
- This allows the TRP to undergo its phase change once introduced to the human body.

Radius of Gyration:

- Radius of gyration of a body about the axis of rotation is defined as the radial distance to a point which would have a moment of inertia the same as the body's actual distribution of mass, if the total mass of the body were concentrated there.

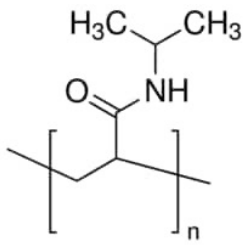


Fig. 1 Chemical Diagram of PNIPAM

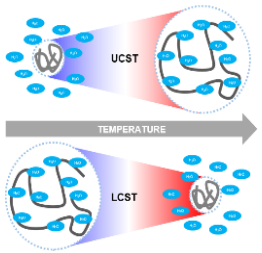


Fig. 2 UCST vs. LCST Diagram

Results:

- All simulations were run through LAMMPS and visualized through VMD

10 Percent Concentration

- An overall downward trend in radius of gyration suggests that PNIPAM with a 10% wt will collapse properly.

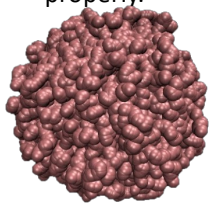


Fig. 3: 10% wt at time 0

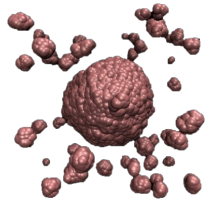


Fig. 4: 10% wt at time 1500

5 Percent Concentration

- An overall downward trend in radius of gyration suggests that PNIPAM with a 5% wt will collapse properly.

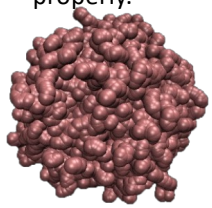


Fig. 5: 5% wt at time 0

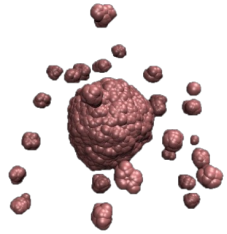


Fig. 6: 5% wt at time 1500

1 Percent Concentration

- An overall upward trend in radius of gyration suggests that PNIPAM with a 1% wt will not collapse properly.



Fig. 7: 1% wt at time 0

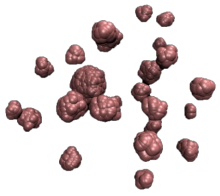


Fig. 8: 1% wt at time 1500

Results (Cont.):

T/Tc vs. Time

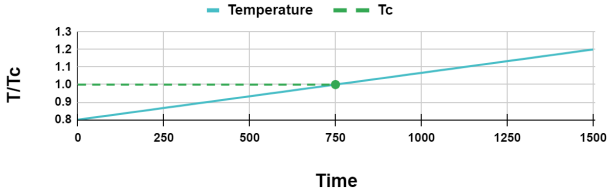


Fig. 9 Graphical Representation of Linear Temperature Increase

Changes in Radius of Gyration

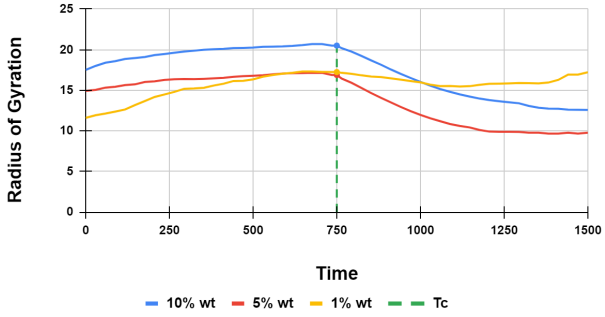


Fig. 10 Graphical Representation of Radius of Gyration Changes

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